

DATA CLEANING & REPORTING AUTOMATION

Objective

The objective of this project is to automate the process of cleaning and reporting sales data using Python. The project demonstrates how data preprocessing techniques can improve data quality and support accurate business analysis.

Tools Used

- Python
- Pandas
- Matplotlib
- Power BI

Methodology

A raw sales dataset containing missing values, duplicate records, and inconsistent information was processed using Python. Duplicate entries were removed, missing values were handled, and the cleaned dataset was exported for further analysis. Automated reports and visualizations were generated to summarize key business metrics.

Key Findings

1. Duplicate records were successfully identified and removed from the dataset.
2. Missing sales values were replaced using the average sales value to preserve data integrity.
3. Missing customer and region information were standardized using placeholder values.
4. Automated reporting reduced manual effort and improved reporting consistency.
5. Visual summaries provided clear insights into regional sales performance and product contributions.

Conclusion

The project demonstrates the importance of data cleaning and automation in analytics workflows. Automated preprocessing and reporting improve efficiency, reduce errors, and enable faster decision-making based on reliable data.